



FIX DROP for US Options

V1.8

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Version	Description	Saved by	Saved on	Status
v1.8				
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v1.6b	FIX Drop for US Options, v1.6b			
v1.6a - 1	Add missing origSeqNo to NewOrderAck msg			
v1.6a	FIX Drop for US Options, v1.6a			
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v1.1	FIX Drop for US Options, v1.1			
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v0.1	FIX Drop for US Options, DRAFT, v0.1			

1 Overview

The FIX Drop for US Options protocol is the application layer message standard used by members to receive client information related to trades executed on the US Options Exchange. FIX Drop for US Options is a text key=value formatted protocol based on the FIX Trading Community FIX 5.0 SP2 specifications.

2 Transport Layer

The FIX Drop for US Options protocol uses the Transmission Control Protocol (TCP) to provide reliable and ordered delivery of messages between clients and servers.

3 Session Layer

The FIX Drop for US Options protocol uses the [FIXT 1.1 specification](#) session messages to establish and maintain a session.

4 Application Layer

FIX Drop for US Options protocol application layer messages are based on the [Financial Information Exchange \(FIX\) Protocol version 5.0](#) and are key=value text encoded.

4.1 ClOrdID

All ClOrdIDs can be up to 20 characters. They can contain only ASCII printable characters.

4.2 Symbology

Upon request via a separate channel, the exchange provides a list of instrument directory messages. Each instrument directory message provides these important fields used to identify a product for order processing:

- OptionsSecurityID, an 8 CHAR field that uniquely identifies an options security on the exchange,
- Symbol (OSIRoot), a six CHAR field that identifies the OSI root symbol,
- MaturityDate, the date of maturity (a full date) of the Local Market (vs. UTC) in YYYYMMDD format,
- PutOrCall, indicates if the option security is a Put or a Call,
- StrikePrice, the strike price for the option security.

The MEMO for Options SBE protocol uses the OptionsSecurityID (custom tag 21035) to identify an option security.

The MEMO for Options FIX protocol allows the use of the OptionsSecurityID (custom tag 21035), or the combination of the Symbol (OSIRoot tag 55), MaturityDate (tag 541), the PutOrCall (tag 201) and the StrikePrice (tags 202) to identify an option security.

Since the MEMX FIX Drop for US Options protocol can drop information from both SBE and the FIX protocols, it provides the OptionsSecurityID (custom tag 21035), Symbol (OSIRoot tag 55), MaturityDate (tag 541), PutOrCall (tag 201) and the StrikePrice (tag 202) in the dropped messages.

5 Message Field Types

5.1 PartyIDSourceType (FIX tag 447)

PartyIDSource is encoded as a char type.

Value	Name
'D'	Proprietary/Custom Code

5.2 PartyRoleType (FIX tag 452)

Identifies the type or role of the PartyID in the Parties repeating group.

PartyRoleType is encoded as a int type.

Value	Name
1	Execution Firm
3	Actionable Identifier
4	Clearing Firm
14	GiveUp Clearing Firm
17	Contra EFID
18	Contra Clearing Firm
24	OptionalOCData
37	Contra MM Sub Account
40	CMTA
41	Contra CMTA
66	Market Maker Sub Account

5.3 CancelRejectReasonCode (FIX tag 102)

Identifies the reason why an order cancellation or replace request was rejected.

Values 0-99 are reserved by the FIX specification. Note that not all of the reserved FIX codes are used.

Values 100-199 are used to denote issues with specific request fields.

Values 200-299 are used to denote issues with the request as a whole.

Values 300-399 are used to denote issues related to risk.

Also note that some codes in this table may not be sent in FIX but are listed here to maintain consistency with the other order entry specifications, or reserved for future use.

CancelRejectReasonCode is encoded as an int type.

Value	Name	Description
0	TooLateToCancel	A cancel or replace request was received after the exchange has closed.
1	UnknownOrder	An order with the specified identifier was not found.
3	OrderAlreadyPendingCancelOrReplace	There is a pending request on this order.

Value	Name	Description
6	DuplicateClOrdID	The supplied ClOrderID was already used on a previous request.
8	PriceExceedsCurrentPriceBand	The price of the replacement order fell outside execution price collar bands.
18	InvalidPriceIncrement	Request Limit Price increment is invalid.
99	Other	This request was rejected for some other reason not listed.
100	MissingSymbol	The request did not supply a Symbol.
101	InvalidSymbol	The request supplied an invalid Symbol.
102	MissingClOrdID	The request did not supply a ClOrderID.
103	InvalidClOrdID	The request supplied an invalid ClOrderID.
104	MissingSide	The request did not supply a side.
105	InvalidSide	The request supplied an invalid side.
106	MissingOrderQty	The request did not supply a quantity.
107	InvalidOrderQty	The request supplied a quantity, but it was set to an invalid value (i.e. 0, larger than exchange max, etc).
108	MissingOrderType	The request did not supply an order type.
109	InvalidOrderType	The request supplied an invalid order type.
110	MissingLimitPrice	The request did not supply a LimitPrice field and was required to due to OrderType requested.
111	InvalidLimitPrice	The request supplied the LimitPrice field but the value was invalid (0, more than the max allowed by the exchange, etc) or not applicable (i.e. the OrderType was set to Market).
112-113	N/A	
114	MissingListSeqNo	The request did not supply the ListSeqNo field.
115	InvalidListSeqNo	The request supplied the ListSeqNo field, but it was set to an invalid value.
116	MissingOrigClOrdID	The request did not supply the OrigClOrderID field.
117	InvalidOrigClOrdID	The request supplied the OrigClOrderID field, but it was too large, contained illegal characters, etc.
118	MissingOrderID	The request did not supply the OrderID field, and was required to (for example, request omitted the OrigClOrderID field and was trying to perform a delegated cancel for an order from another port).
119	InvalidOrderID	The request supplied the OrderID field, but it was too large, contained illegal characters, etc.
120	MissingSendingTime	The request did not supply the sending time
121	InvalidSendingTime	The request supplied sendingTime, but it was set to an invalid value.
122-199	N/A	Reserved.
200	Halted	The symbol supplied in the request is currently halted.

Value	Name	Description
201	FirmDisabled	The request was rejected because the firm sending the order has been disabled.
202	EFIDDisabled	The request was rejected because the EFID used in the order has been disabled.
203	AccountDisabled	The request was rejected because the account sending the order has been disabled.
204	UnsupportedOrdTypeChange	Change of OrdType in replace request not supported.
205	UnsupportedSideChange	Change of side in replace request not supported.
206	SymbolMismatch	Symbol provided on a replace request does not match the Symbol on the original order associated with the provided OrigClOrdID or OrderID.
207	OrigOrderIDMismatch	OrigClOrdID and OrderID were both provided, but the provided OrderID does not match the actual OrderID associated with the provided OrigClOrdID.
208	UnsupportedOrderPropertyChange	Attempting to change an immutable order property.
209-299	N/A	Reserved.
300	NonTestSymbolsBlocked	The replace was for a non-test symbol and the port is configured in test-only mode (exchange configuration).
301	NotionalValueExceedsThreshold	The notional value of the replace exceeded an exchange configured maximum.
302	MaxNotionalValuePerOrderRiskRuleViolated	The notional value of the replace violated the firm's max notional value per order risk rule.
303	BlockNonTestSymbolRiskRuleViolated	The replace violated the firm's test symbol trading risk rule.
304	MaxContractsPerOrderRiskRuleViolated	The replace breached the firm's max contracts per order risk rule.
305	NBBOWidthExceedsThreshold	The width of the NBBO is greater than the exchange specified threshold.
306	ExchangePriceValueCollarRiskRuleViolated	A replace Limit Order to buy with a price that is more than an Exchange specified amount above the NBO or an incoming replace Limit Order to sell with a price that is less than an Exchange specified amount below the NBB
307	ExecutionPriceHigherThanStrikePrice	Replace Market Order or the remaining portion of a Market Order to buy a put that would execute in whole or in part at a price higher than or equal to the strike price of the option or an incoming Limit Order to buy a put that would execute at a price higher than or equal to the strike price of the option.
308	MarketBuyWhenNBOIsZero	Replace Market Order to buy in an open series when the NBO is zero.
309	MarketSellWhenNBBGreaterThanThreshold	Replace Market Order to sell in an open series when the NBB is greater than \$0.50.

Value	Name	Description
310	QuotePriceOutsideExchangeThreshold	Bulk quote replace canceled, bid with a price that is more than an Exchange specified amount above the NBO and cancel an incoming bulk replace offer with a price that is less than an Exchange specified amount below the NBB.
311	N/A	
312	NumContractsExecutedExceedsThreshold	The number of contracts executed exceeds firm specified threshold.
313	NotionalValueOfExecutionsExceedsThreshold	The notional value of contracts executions exceeds firm specified threshold.
314	CountOfExecutionsExceedsThreshold	The count of contracts executed exceeds firm specified threshold.
315	OutstandingPercentageThresholdExceeded	The number of contracts executed as a percentage of the number of contracts outstanding within an Exchange configurable time period or for the trading day has been exceeded.
316	TripsThresholdExceeded	The number of times a firm reaches the limits in: - NumContractsExecutedExceedsThreshold - NotionalValueOfExecutionsExceedsThreshold - CountOfExecutionsExceedsThreshold - OutstandingPercentageThresholdExceeded exceeds a threshold.
317	ISOOrdersNotAllowed	ISO marked orders are disabled.
318	MarketIsCrossed	Order not allowed while market is crossed.
319	ActiveRiskBreach	Order not allowed because of an active breach.
320	ManualRiskBreach	Order not allowed because of a manual breach.
321	GrossNotionalValueExceedsThreshold	Order not allowed because it surpassed the gross notional risk threshold.
322	NetNotionalValueExceedsThreshold	Order not allowed because it surpassed the net notional risk threshold.
323	DuplicateOrderThresholdExceeded	Order not allowed because it surpassed the duplicate order risk threshold.
324	OrderRateThresholdExceeded	Order not allowed because it surpassed the order rate risk threshold.
325	MassCancelLockoutInEffect	Order not allowed because a mass cancel lockout is in effect.
326	MarketOrderGrossNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order gross notional risk threshold.
327	MarketOrderNetNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order net notional risk threshold.
328	LimitOrderFatFingerCheck	Limit Order not allowed because it failed fat finger check

Value	Name	Description
329	BulkQuoteFatFingerCheck	Bulk Quote not allowed because it failed fat finger check

5.4 CancelReasonCode (MEMX custom FIX tag 21004)

CancelReasonCode represents the reason the order was canceled by the system.

CancelReasonCode is encoded as an int type.

Value	Name	Description
0	Other	This order was canceled for some other reason not listed.
1	UserRequestedCancel	The client sent a OrderCancelRequest or OrderMassCancelRequest for this order.
5	ExecutionPriceCollar	The price of the order fell outside execution price collar bands
6	Halted	The market on the order's security was halted.
7	ExchangeSupervisory	Operational or supervisory actions taken by the Exchange resulted in the cancellation of this order.
8	OrderExpired	The order's supplied expiration time passed.
10	MatchTradePrevention	This or another associated order's specified match trade prevention behavior triggered the cancellation of this order.
13	ParticipantDisconnect	The participant directed that their orders should be canceled when the trading system detects a disconnection, and the participant disconnected.
14	OrderNotBookable	Order is not of bookable type (this may include market orders, IOC, etc)
17	FirmDisabled	Order was canceled because the firm was disabled
18	EFIDDisabled	Order was canceled because the EFID was disabled
19	AccountDisabled	Order was canceled because the Account was disabled
20, 21	N/A	Reserved
22	QuoteUpdateRejectedOrigQuoteCanceled	A quote update for an existing quote was rejected, the original quote was canceled.
23	ActiveRiskBreach	Canceled because an active risk rule was breached.
24	ManualRiskBreach	Canceled because of a manual breach.
25	NumContractsExecutedExceedsThreshold	The number of contracts executed exceeds firm specified threshold.
26	NotionalValueOfExecutionsExceedsThreshold	The notional value of the order violated the firm's max notional value per order risk rule.
27	CountOfExecutionsExceedsThreshold	The count of contracts executed exceeds firm specified threshold.

Value	Name	Description
28	OutstandingPercentageThresholdExceeded	The number of contracts executed as a percentage of the number of contracts outstanding within an Exchange configurable time period or for the trading day has been exceeded.
29	TripsThresholdExceeded	The number of times a firm reaches the limits in: • NumContractsExecutedExceedsThreshold • NotionalValueOfExecutionsExceedsThreshold • CountOfExecutionsExceedsThreshold • OutstandingPercentageThresholdExceeded exceeds a threshold.
30	GrossNotionalValueExceedsThreshold	Order not allowed because it surpassed the gross notional risk threshold.
31	NetNotionalValueExceedsThreshold	Order not allowed because it surpassed the net notional risk threshold.
32	DuplicateOrderThresholdExceeded	Order not allowed because it surpassed the duplicate order risk threshold.
33	OrderRateThresholdExceeded	Order not allowed because it surpassed the order rate risk threshold.
34	MarketOrderGrossNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order gross notional risk threshold.
35	MarketOrderNetNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order net notional risk threshold.
36	MarketOrderNBBOWidthExceeded	Market Order not allowed because it surpassed the market order NBBO width
37	MarketOrderNoNationalBestBid	Market Order not allowed because there was no national best bid
38	MarketOrderNoNationalBestOffer	Market Order not allowed because there was no national best offer
39	DrillThroughPriceExceeded	Order not allowed because it surpassed drill through price
40	OrderWouldLockOrCross	Order not allowed because it would lock or cross
41	MassCancelRequest	Cancel was the result of a mass cancel request
42	LockedOrCrossedMarket	Cancel was due to the order having a RepriceBehavior specified as RepriceLockCancelCross and the Away Market Quotes being crossed

5.5 CxlRejResponseToType (FIX tag 434)

CxlRejResponseToType identifies the request type that a cancel reject was issued for.

CxlRejResponseToType is encoded as char type.

Value	Name	Description
'1'	OrderCancelRequest	This response is for a cancel request.
'2'	OrderCancelReplaceRequest	This response is for a replace request.

5.6 ExchangeCode (FIX tag 30)

ExchangeCode (MIC code) indicates the market of execution for last fill, or an indication of the market where an order was routed. MIC code are maintained here [Market identifier codes | ISO20022](#).

ExchangeCode is encoded as string.

Value	Market Name
AMXO	NYSE AMEX OPTIONS
XBOX	BOX OPTIONS EXCHANGE
XCBO	CBOE GLOBAL MARKETS INC.
EMLD	MIAX EMERALD, LLC
EDGO	CBOE EDGX OPTIONS EXCHANGE
GMNI	ISE GEMINI EXCHANGE
XISX	INTERNATIONAL SECURITIES EXCHANGE, LLC
MCRY	ISE MERCURY, LLC
XMIO	MIAMI INTERNATIONAL SECURITIES EXCHANGE
ARCO	NYSE ARCA OPTIONS
MPRL	MIAX PEARL, LLC
XNDQ	NASDAQ OPTIONS MARKET
XBXO	NASDAQ OMX BX OPTIONS
MXOP	MEMX OPTIONS EXCHANGE
C2OX	CBOE C2 OPTIONS EXCHANGE
XPHO	PHILADELPHIA OPTIONS EXCHANGE
BATO	CBOE BZX OPTIONS EXCHANGE

5.7 ExecInst (FIX tag 18)

The ExecInst field provides instructions for order handling on the exchange.

ExecInst is encoded as a MultipleCharValue type. Encoded as a string with one or more values below, separated by spaces.

Value	Name	Description
'6'	ParticipateDoNotInitiate	Post-Only - order will not attempt to remove liquidity on entry.
'f'	IntermarketSweep	IntermarketSweep
'h'	ExternalRoutingNotAllowed	The order cannot be routed away.

5.8 ExecRestatementType (FIX tag 378)

Defines the reason for a restatement.

ExecRestatementType is encoded as an int type.

Value	Name	Description
3	OrderReprice	The order's booking price was updated.
5	MatchTradePrevention	The order quantity was reduced to prevent a wash trade.

Value	Name	Description
99	Other	More information is supplied in tag 21022 ExtendedExecRestatementReason.
100	PostTradeAllocation	The order details were changed in post trade.

5.9 ExtendedRestatementReasonType (Custom FIX tag 21022)

Defines the extended reason for a restatement.

ExtendedRestatementReasonType is as an int type.

Value	Name	Description
0	None	No additional information.
1, 2	Reserved	Reserved for future use.
3	MatchTrade-CancelNewest	Restatement due to MatchTradePrevention - CancelNewest
4	MatchTrade-CancelOldest	Restatement due to MatchTradePrevention - CancelOldest
6	MatchTrade-CancelBoth	Restatement due to MatchTradePrevention - CancelBoth

5.10 PositionEffectType (FIX tag 77)

Indicates whether the resulting position after a trade should be an opening position or closing position.

PositionEffectType is a char type.

Value	Name
'O'	Open
'C'	Close

5.11 AllocTransType (FIX tag 71)

The AllocTransType Identifies an allocation transaction type.

AllocTransType is encoded as an int type.

Value	Name	Description
0	New	Replace existing trade with allocation. This is the first allocation on a trade, and does not require a RefAllocID to be supplied. The original trade must be referenced in the Execution Allocations Group.
1	Replace	Replacing existing allocations with new allocations. This is any subsequent allocation and clients should use the RefAllocID field to match a previously reported allocation by specifying its IndividualAllocID value.
2	Cancel	Canceling existing allocations. This is done by the exchange when the original trade is canceled or corrected. Will reference the individual allocation via the RefAllocID field.

5.12 AllocPositionEffectType (FIX tag 1047)

Indicates whether the resulting position for an allocation should be an opening or closing position.

AllocPositionEffectType is a CHAR value.

Value	Name
'O'	Open
'C'	Close

5.13 AllocType (FIX tag 626)

The AllocType Identifies an allocation type.

AllocType is encoded as an int type.

Value	Name	Description
100	PostTradeEdit	The allocation was a post trade edit. (This is a custom value defined by MEMX)

5.14 OrderRejectReasonCode (FIX tag 103)

Identifies the reason why a new order was rejected by the system.

Values 0-99 are reserved by the FIX specification. Note that not all of the reserved FIX codes are used.

Values 100-199 are used to denote issues with specific request fields.

Values 200-299 are used to denote issues with the request as a whole.

Values 300-399 are used to denote issues related to risk.

Also note that some codes in this table may not be sent in FIX but are listed here to maintain consistency with the other order entry specifications, or reserved for future use.

OrderRejectReasonCode is encoded as an int type.

Value	Name	Description
1	UnknownSymbol	The order contained a valid symbol identifier, but the symbol is not currently traded on this port or exchange.
2	ExchangeClosed	The exchange is closed and not accepting orders at this time.
6	DuplicateOrder	The user entered a duplicate order identifier (ClOrdID).
16	PriceExceedsCurrentPriceBand	The price of the order fell outside execution price collar bands.
18	InvalidPriceIncrement	The order contained an invalid limit price increment.
27	MaxNotionalValuePerOrderRiskRuleBreach	The order breached the firm's max notional value per order risk rule.
99	Other	This order was rejected for some other reason not listed.
100	MissingSymbol	The order did not supply the Symbol.
101	InvalidSymbol	The order had an. invalid Symbol.
102	MissingClOrdID	The order did not supply the ClOrdID.

Value	Name	Description
103	InvalidClOrdID	A ClOrdID was present but contained illegal characters.
104	MissingSide	The order did not supply the side.
105	InvalidSide	The order contained an invalid side.
106	MissingOrderQty	The order did not supply the quantity.
107	InvalidOrderQty	The order contained an invalid order quantity.
108	MissingOrderType	The order did not supply the order type.
109	InvalidOrderType	The order contained an invalid order type.
110	MissingTimeInForce	The order did not supply the time in force.
111	InvalidTimeInForce	The order contained an invalid time in force.
112	MissingTradingCapacity	The order did not supply the trading capacity.
113	InvalidTradingCapacity	The order contained an invalid trading capacity.
114	MissingExecInst	The order did not supply the execution instructions field.
115	InvalidExecInst	The order contained an invalid execution instruction.
116, 117	N/A	
118	MissingLimitPrice	The order did not supply the limit price.
119	InvalidLimitPrice	The order contained an invalid limit price.
120 - 123	N/A	
124	MissingMatchTradePreventionType	The order supplied the MTPGroupID field but did not supply the MatchTradePreventionType field.
125	InvalidMatchTradePreventionType	The order contained an invalid MatchTradePreventionType.
126	MissingCancelGroupID	The order did not supply the CancelGroupID field (used when a port is configured with a fixed set of allowable CancelGroupIDs).
127	InvalidCancelGroupID	The order contained an invalid CancelGroupID (used when a port is configured with a fixed set of allowable CancelGroupIDs).
128	MissingMTPGroupID	The order did not supply the MTPGroupID field (used when a port is configured with a fixed set of allowable MTPGroupIDs).
129	InvalidMTPGroupID	The order contained an invalid MTPGroupID (used when a port is configured with a fixed set of allowable MTPGroupIDs).
130- 133	N/A	Reserved.
134	MissingRiskGroupID	The order did not supply the RiskGroupID field (used when a port is configured with a fixed set of allowable RiskGroupIDs).
135	InvalidRiskGroupID	The order contained an invalid RiskGroupID (used when a port is configured with a fixed set of allowable RiskGroupIDs).
136	MissingEFID	The order did not supply the EFID field.

Value	Name	Description
137	InvalidEFID	The order contained an invalid EFID.
138	MissingListSeqNo	The order did not supply the ListSeqNo field.
139	InvalidListSeqNo	The order contained an invalid ListSeqNo.
140	QuotesHaveDifferentUnderliers	The order contained quotes for different underliers.
141	TwoSidedQuotesCross	The order contained a two sided quote's where bidPx and offerPx cross.
142	MissingPositionEffect	The order did not supply the PositionEffect field.
143	InvalidPositionEffect	The order contained an invalid PositionEffect.
144	MissingRepriceBehaviorType	The order did not supply the RepriceBehaviorType field.
145	InvalidRepriceBehaviorType	The order contained an invalid RepriceBehaviorType.
146	MissingRepriceFrequencyType	The order did not supply the RepriceFrequencyType field.
147	InvalidRepriceFrequencyType	The order contained an invalid RepriceFrequencyType.
148	MissingPartyRoleType	The order did not supply the PartyRoleType field.
149	InvalidPartyRoleType	The order contained an invalid PartyRoleType.
150	MissingPartyID	The order did not supply the PartyID field.
151	InvalidPartyID	The order contained an invalid PartyID.
152	MissingPartyIDSource	The order did not supply the PartyIDSource field.
153	InvalidPartyIDSource	The order contained an invalid PartyIDSource.
154	UnderlyingSymbolNotOpen	Underlying symbol has not been open for trading
155	MissingSendingTime	The order did not supply sendingTime field
156	InvalidSendingTime	The order contained invalid sending time field
157	MissingSubAccount	The order did not supply subAccount
158	InvalidSubAccount	The order contained an invalid subAccount
159	MissingGiveUp	The order did not supply giveUp field
160	InvalidGiveUp	The order contained an invalid giveUp
161	MissingCMTA	The order did not supply CMTA field
162	InvalidCMTA	The order contained an invalid CMTA
163	MissingActionableID	The order did not supply actionableID field
164	InvalidActionableID	The order contained an invalid actionableID
165	MissingOptionalOCData	The order did not supply optionalOCData field
166	InvalidOptionalOCData	The order contained an invalid optionalOCData
167-199	N/A	Reserved.
200	Halted	The symbol supplied in the order is currently halted.
201	FirmDisabled	The order was rejected because the firm sending the order has been disabled.

Value	Name	Description
202	EFIDDisabled	The order was rejected because the EFID used in the order has been disabled.
203	AccountDisabled	The order was rejected because the account sending the order has been disabled.
204	InvalidModifierForOrderType	The order contained one or more modifiers that are not compatible with the order type provided.
205	InvalidTimeInForceForOrderType	The order contained a time in force that is not compatible with the order type provided.
206	N/A	Reserved.
207	PostOnlyNotAllowed	Post-only behavior is not allowed with one or more of the supplied order directives.
208	QuoteModifyRejected	Canceled resting bulk quote if an incoming bulk message to modify such quote with the same EFID is rejected or canceled by the Trading System.
209	QuotingDisabled	Bulk quote sent by a non-market maker.
210	InvalidQuoteComponentCount	Empty quote or a quote exceeding max components.
211-299	N/A	Reserved.
300	NonTestSymbolsBlocked	The order was for a non-test symbol and the port is configured in test-only mode (exchange configuration).
301	NotionalValueExceedsThreshold	The notional value of the order exceeded an exchange configured maximum.
302	MaxNotionalValuePerOrderRiskRuleViolated	The notional value of the order violated the firm's max notional value per order risk rule.
303	BlockNonTestSymbolRiskRuleViolated	The order violated the firm's test symbol trading risk rule.
304	MaxContractsPerOrderRiskRuleViolated	The order breached the firm's max contracts per order risk rule.
305	NBBOWidthExceedsThreshold	The width of the NBBO is greater than the exchange specified threshold.
306	ExchangePriceValueCollarRiskRuleViolated	An incoming Limit Order to buy with a price that is more than an Exchange specified amount above the NBO or an incoming Limit Order to sell with a price that is less than an Exchange specified amount below the NBB
307	ExecutionPriceHigherThanStrikePrice	Market Order or the remaining portion of a Market Order to buy a put that would execute in whole or in part at a price higher than or equal to the strike price of the option or an incoming Limit Order to buy a put that would execute at a price higher than or equal to the strike price of the option.
308	MarketBuyWhenNBOIsZero	Market Order to buy in an open series when the NBO is zero.
309	MarketSellWhenNBBGreaterThanThreshold	An incoming Market Order to sell in an open series when the NBB is greater than \$0.50.
310	QuotePriceOutsideExchangeThreshold	Bulk quote canceled, bid with a price that is more than an Exchange specified amount above the NBO and cancel an incoming bulk

Value	Name	Description
		quote message offer with a price that is less than an Exchange specified amount below the NBB.
311	N/A	
312	NumContractsExecutedExceedsThreshold	The number of contracts executed exceeds firm specified threshold.
313	NotionalValueOfExecutionsExceedsThreshold	The notional value of contracts executions exceeds firm specified threshold.
314	CountOfExecutionsExceedsThreshold	The count of contracts executed exceeds firm specified threshold.
315	OutstandingPercentageThresholdExceeded	The number of contracts executed as a percentage of the number of contracts outstanding within an Exchange configurable time period or for the trading day has been exceeded.
316	TripsThresholdExceeded	The number of times a firm reaches the limits in: • NumContractsExecutedExceedsThreshold • NotionalValueOfExecutionsExceedsThreshold • CountOfExecutionsExceedsThreshold • OutstandingPercentageThresholdExceeded exceeds a threshold.
317	ISOOrdersNotAllowed	ISO marked orders at disabled.
318	MarketIsCrossed	Order not allowed while market is crossed.
319	ActiveRiskBreach	Active risk controls were breached for the underlying.
320	ManualRiskBreach	Order not allowed because of a manual breach.
321	GrossNotionalValueExceedsThreshold	Order not allowed because it surpassed the gross notional risk threshold.
322	NetNotionalValueExceedsThreshold	Order not allowed because it surpassed the net notional risk threshold.
323	DuplicateOrderThresholdExceeded	Order not allowed because it surpassed the duplicate order risk threshold.
324	OrderRateThresholdExceeded	Order not allowed because it surpassed the order rate risk threshold.
325	MassCancelLockoutInEffect	Order not allowed because a mass cancel lockout is in effect.
326	MarketOrderGrossNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order gross notional risk threshold.
327	MarketOrderNetNotionalValueExceedsThreshold	Market Order not allowed because it surpassed the market order net notional risk threshold.
328	LimitOrderFatFingerCheck	Limit Order not allowed because it failed fat finger check
329	BulkQuoteFatFingerCheck	Bulk Quote not allowed because it failed fat finger check
330	MarketOrdersNotAllowed	Market orders are disabled

Value	Name	Description
331	UnderlierOnRestrictedList	Order not allowed because underlier is on restricted list

5.15 OrdType (FIX tag 40)

Indicates the type of an order.

OrdType is encoded as a char type.

Value	Name
'1'	Market
'2'	Limit

5.16 LastLiquidityInd (FIX tag 851)

Indicates the liquidity code for this execution.

LastLiquidityInd is encoded as an int type.

Value	Name	Description
1	Added	The order added displayed liquidity to the Exchange book.
2	Removed	The order removed liquidity from the Exchange book.

5.17 TradingCapacityType (FIX tag 1815)

Capacity of the order.

The TradingCapacity is encoded as an int type.

Value	Name
1	Customer
2	Professional Customer
3	Broker Dealer
4	Broker Dealer Customer
5	Firm
6	Market Maker
7	Away Market Maker

5.18 PutOrCallType (FIX tag 201)

Defines the option as a Put or a Call.

PutOrCallType is encoded as an int type.

Value	Name
0	Put
1	Call

5.19 RepriceFrequencyType (MEMX custom FIX tag 21020)

Defines the type of reprice used when marking an order for repricing.

RepriceFrequencyType is an int type.

Value	Name	Description
0	SingleReprice	An order will be repriced upon entry.
1	ContinuousReprice	An order will continuously be repriced after being booked.
2	None	No repricing will be done.

5.20 RepriceBehaviorType (MEMX custom FIX tag 21021)

Defines the market conditions for which to apply repricing for.

RepriceBehaviorType is an int value.

Value	Name	Description
1	RepriceLockCancelCross	An order will only be repriced if the market is locked, cancel if market is crossed.
2	RepriceLockRepriceCross	An order will be repriced if the market is locked or crossed.

5.21 MatchTradePreventionType (MEMX custom FIX tag 21001)

Defines the self trade operation which will be used if the exchange match-trade-prevention functionality is triggered on this order.

If this order would execute against an opposite side resting interest with the same match-trade prevention identifier, the behavior specified by this value will be triggered.

MatchTradePreventionType is encoded as an int type.

Value	Name	Description
0	CancelNewest	If triggered, this order (the incoming order) will be canceled.
1	CancelOldest	If triggered, the resting order will be canceled.
2	N/A	Reserved value.
3	CancelBoth	If triggered, both the incoming and the resting orders will be canceled.

5.22 SideType (FIX tag 54)

Defines the side of an order.

SideType is a char type.

Value	Name
'1'	Buy
'2'	Sell
'B'	As Defined - Note this is not a valid Client to Exchange SideType value. It is only sent from the Exchange to the Client. It can be used for the Side field in an Rejected Order message when an entire bulk quote is rejected.

5.23 MtpGroupIDType (FIX tag 2362)

This field uniquely identifies the match trade prevention group. Several of the values are reserved values while other values are treated as custom values and are at the discretion of the client.

MtpGroupID is encoded as an int type.

Value	Name	Description
0	Firm Scope	Prevents orders from the same firm trading with each other
1	EFID Scope	Prevents orders from the same EFID trading with each other
2	Account Scope	Prevents orders from the same account trading with each other
3-65534	Custom Scope	Prevents orders with matching MtpGroupID across the firm from trading with each other

5.24 TimeInForceType (FIX tag 59)

Defines the time during which an order is eligible for execution.

TimeInForceType is encoded as a char type.

Value	Name	Description
'0'	Day	The order is entered for execution during the pre-market and market periods.
'3'	Immediate Or Cancel (IOC)	The order shall be partially or completely executed immediately, or canceled outright.

6 Headers and Trailers

All messages begin with a *Standard Header* and must end with a *Standard Trailer*.

6.1 Standard Header

The *Standard Header* must contain the required fields listed and can contain any of the non-required listed fields:

Tag	Field Name	Req'd	Meaning/Values
8	BeginString	Y	"FIXT.1.1" This must be the first field in the message.
9	BodyLength	Y	The length of the FIX message. This must be the second field in the message.
35	MsgType	Y	The type of message. This must be the third field in the message.
49	SenderCompID	Y	The sender's id, assigned by at on-boarding time.
56	TargetCompID	Y	"MXOP"
34	MsgSeqNum	Y	The message sequence number.
52	SendingTime	Y	The time of message transmission in Universal Time coordinated. See UTCTimestamp in FIX 5.0 SP2 specification.
43	PossDupFlag	N	Always required for retransmitted messages, whether prompted by the sending system or as the result of a resend request.
97	PossResend	N	Required when message may be duplicate of another message sent under a different sequence number.
122	OrigSendingTime	N	Required for message resent as a result of a ResendRequest. If data is not available set to same value as SendingTime.

6.2 Standard Trailer

The *Standard Trailer* must contain the CheckSum field.

Tag	Field Name	Req'd	Meaning/Values
10	Checksum	Y	The message checksum. This must be the last field in the message.

7 Messages

This section defines the messages that make up the protocol. For each message, it lists the fields in the message by tag id and name, whether the field is required and acceptable values or description of the field.

7.1 Session Messages

This section defines the session level messages supported.

7.2 Heartbeat (Sent by exchange and client)

The Heartbeat message confirms the status of the communication link.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 0
112	TestReqID	N	Identifier included in Test Request message to be returned in resulting Heartbeat. (Max Length: 64 characters)
	Standard Trailer	Y	

7.3 Logon

The logon message is sent by the client to establish a connection to the exchange.

A logon message must be the first message sent by the application initiating a FIX session.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = A
98	EncryptMethod	Y	Required to be 0.
108	HeartBtInt	Y	Heartbeat interval, in seconds. An interval of 0 indicates there are no heartbeats. The maximum supported interval is 90 seconds.
141	ResetSeqNumFlag	N	FIX Drop Copy will reject logon if this flag is 'Y'. Market Operations can reset the inbound and outbound sequence numbers on request.
1137	DefaultAppVerID	Y	Required to be 9. ("FIX50SP2")
1408	DefaultCstmAppVerID	Y	The schema version determined on port creation and displayed in User Portal.
	Standard Trailer	Y	

Note: If the DefaultCstmAppVerID (tag 1408) is invalid, the client connection is dropped.

7.4 Logout

The logout message initiates or confirms the termination of a FIX session.

This message may be sent by the exchange or by the client.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 5
58	Text	N	Text explaining reason for logout. (Max Length: 128 characters)
	Standard Trailer	Y	

7.5 Reject

The reject message should be issued when a message is received but cannot be properly processed due to a session-level rule violation.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 3
45	RefSeqNum	Y	Reference message sequence number.
371	RefTagID	N	The tag number of the FIX field that was rejected.
372	RefMsgType	N	The MsgType of the FIX message that was rejected.
58	Text	N	Explanatory reason. (Max Length: 128 characters)
	Standard Trailer	Y	

7.6 ResendRequest

The resend request is sent by the receiving application to initiate the retransmission of messages.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 2
7	BeginSeqNo	Y	Message sequence number of first message in range to be resent.
16	EndSeqNo	Y	Message sequence number of last message in range to be resent.
	Standard Trailer	Y	

7.7 SequenceReset

The sequence reset message is used to reset the incoming sequence number on the opposing side.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 4
123	GapFillFlag	N	Indicates that this message is replacing administrative or application messages which will not be resent.
36	NewSeqNo	Y	New sequence number.
	Standard Trailer	Y	

7.8 TestRequest

The test request message forces a heartbeat from the opposing application.

The opposing application responds to the Test Request with a Heartbeat containing the TestReqID.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 1
112	TestReqID	Y	Identifier included in Test Request message to be returned in resulting Heartbeat. (Max Length: 64 characters)
	Standard Trailer	Y	

8 Session Management

The client establishes a session by opening a TCP connection to the exchange FIX Drop port (server) and sending a FIX logon message with the appropriate credentials and MsgSeqNum. The SenderCompID and TargetCompID will be provisioned by the exchange and will be verified by the server at Logon. If the Logon message is accepted, the server will send a logon response. If the MsgSeqNum on the inbound logon is less than expected by the server, the logon will be rejected.

If the MsgSeqNum of the inbound Logon message is greater than expected, the server will reply with a Logon, and shall send a ResendRequest for the missing messages. The client application should gap fill any session level messages (see [FIXT 1.1 specification](#) Message Recovery), but may opt to issue sequence reset messages with the GapFillFlag set.

There are no client to server business level messages specified in this Drop Copy API specification. When the client application receives the Logon response, it should compare the MsgSeqNum to the expected server sequence number. If a gap is detected, it should send a ResendRequest message to the server. The server will resend any missing business level messages with the PossDup flag set. Instead of sending session-level messages during a backfill, the server will issue sequence reset messages with the GapFillFlag set.

When the client terminates a session it should send a Logout message and wait for the Logout response from the server before closing the connection. This procedure facilitates gap detection and processing before the connection is closed.

At the end of a logical trading session, the FIX inbound and outbound sequence numbers are set to 1. All inbound FIX messages will be rejected if a FIX session has not been established. If the server receives a message with a MsgSeqNum less than the expected client sequence number, and that message does not have the PossDup flag set, the server will disconnect the client. In the event of a failover to a backup port, the backup port will maintain the sequence numbers of the primary, and it will have the ability to replay any messages that the primary port sent.

9 Application Messages From Exchange to Client

9.1 ExecutionReport

9.1.1 Trade

This execution report is sent when a trade occurs.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	Unique identifier for an Order as assigned by the client.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	F - Trade (partial fill or fill).
39	OrdStatus	Y	1 - Partially filled 2 - Filled
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
67	ListSeqNo	Y	As stated in Bulk Quote request or 0 for NewOrderSingle request.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the order.
32	LastQty	Y	Quantity bought/sold on this (last) fill.
31	LastPx	Y	Price of this (last) fill.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
40	OrdType	Y	As stated on the previous order.
1815	TradingCapacity	Y	As stated in the order.
77	PositionEffect	N	As stated in the order.
60	TransactTime	Y	The time at which the transaction occurred.
880	TrdMatchID	Y	Identifier assigned to the Trade by the matching system.
851	LastLiquidityInd	Y	Indicator denoting whether the referenced order removed liquidity from or added liquidity to the Exchange book.
30	LastMkt	Y	Market of execution for last fill, or an indication of the market where an order was routed. See ExchangeCode (FIX tag 30)
1	Account	Y	Account associated to the order.
21036	ContraTradingCapacity	Y	TradingCapacity of the contra order.
167	SecurityType	Y	Indicates type of security.
453	NoPartyIDs	Y	The number of parties in the following Parties Group
→448	PartyID	Y	

Tag	Field Name	Req'd	Meaning/Values
→447	PartyIDSource	Y	
→452	PartyRole	Y	
	Standard Trailer	Y	

9.1.2 Trade Cancel/Correct

The Trade Cancel/Correct execution report is sent to indicate that the price or quantity of a trade has been corrected. A trade cancel execution report is sent to bust a previous trade.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	Unique identifier for an Order as assigned by the client.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
19	ExecRefID	Y	The ExecID of the trade being broken.
150	ExecType	Y	G - TradeCorrect H - TradeCancel
39	OrdStatus	Y	1 - Partially filled 2 - Filled 4 - Canceled
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
44	Price	N	As stated on the order
38	OrderQty	Y	As stated in the order.
31	LastPx	N	The corrected price. Required if Trade Correct.
32	LastQty	N	The corrected quantity. Required if Trade Correct.
6	AvgPx	Y	The calculated average price of all fills for the executed order.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
880	TrdMatchID	Y	Identifier assigned to the Trade by the matching system.
	Standard Trailer	Y	

9.2 Order Drop Execution Reports

FIX Drop Copy by default will always send Trade execution reports. If clients opt in for Order Drop, the port will also disseminate the order entry execution reports listed below.

9.2.1 New Order Acknowledge

The New Order Acknowledge execution report is a response to the NewOrderSingle message in the event the order was accepted by the exchange. This message echos back to the requester the accepted state of the order.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	As stated in the order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	0 - New
39	OrdStatus	Y	0 - New
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for further execution.
14	CumQty	Y	0
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
40	OrdType	Y	As stated in the order.
59	TimeInForce	Y	As stated in the order.
1815	TradingCapacity	Y	As stated in the order.
21020	RepriceFrequency	N	As stated in the order.
21021	RepriceBehavior	N	As stated in the order.
18	ExecInst	Y	As stated in the order.
21001	MatchTradePreventionType	N	As stated in the order. This is a custom FIX tag.
2362	MTPGroupID	N	As stated in the order.
21000	CancelGroupID	N	As stated in the order.
21005	RiskGroupID	N	As stated in the order. This is a custom FIX tag.
60	TransactTime	Y	The time at which the transaction occurred. UTC timestamp since epoch with millisecond precision.
77	PositionEffect	N	As stated in the order.
1	Account	Y	Account associated to the order.
67	ListSeqNo	Y	As stated in the order. Always 0 for NewOrderSingle request.
453	NoPartyIDs	Y	The number of parties in the following Parties Group
→448	PartyID	Y	
→447	PartyIDSource	Y	
→452	PartyRole	Y	
	Standard Trailer	Y	

9.2.2 Rejected Order

The Rejected Order execution report is sent in response to a NewOrderSingle of Bulk Quote message that was rejected by the exchange.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8

11	ClOrdID	Y	As stated in the order.
37	OrderID	Y	Unique identifier for an Order assigned by the exchange, if available.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	8 - Rejected
39	OrdStatus	Y	8 - Rejected
21035	OptionsSecurityID	N	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	N	The OSI Root for the options security.
541	MaturityDate	N	Date of maturity in YYYYMMDD format.
201	PutOrCall	N	Indicates whether the option security is a put or a call.
202	StrikePrice	N	The Strike Price for the options security.
54	Side	Y	As stated in the order. If the order was a Bulk Quote this field will have the value 'B' = "As Defined".
151	LeavesQty	Y	0
14	CumQty	Y	0
44	Price	N	As stated in the order.
38	OrderQty	Y	As stated in the order.
40	OrdType	Y	As stated in the order.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
103	OrdRejReason	Y	Reason code for order rejection.
1	Account	Y	Account associated to the order.
	Standard Trailer	Y	

9.2.3 Replaced

The Replaced execution report is sent when an order has been replaced.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	As stated in the replace request. Will be applied as the new ClOrdID for this order going forward.
41	OrigClOrdID	Y	The ClOrdID associated with the replaced order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	5 - Replaced
39	OrdStatus	Y	0 - New 1 - PartiallyFilled
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the replace request.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.

Tag	Field Name	Req'd	Meaning/Values
44	Price	N	As stated in the replace request.
38	OrderQty	Y	As stated in the replace request.
40	OrdType	Y	As stated in the replace request.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
1	Account	Y	Account associated to the order.
67	ListSeqNo	Y	As stated in the cancel request
21023	OrigListSeqNo	Y	Original list sequence number
	Standard Trailer	Y	

9.2.4 Canceled

The Canceled execution report is sent when an order has been canceled. This may be because of a client requested cancel, a replace, or a cancel initiated by the exchange. Orders that are canceled by the exchange because of Time in Force instructions shall have an ExecType and OrdStatus as Expired. All other cases shall have ExecType and OrdStatus as Canceled.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	As stated in the cancel request or the ClOrdID of the order if an unsolicited cancel.
41	OrigClOrdID	Y	The ClOrdID associated with the canceled order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	4 - Canceled C - Expired
39	OrdStatus	Y	4 - Canceled C - Expired
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the cancel request.
151	LeavesQty	Y	0
14	CumQty	Y	Total quantity filled on the order.
44	Price	N	As stated on the order.
38	OrderQty	Y	As stated on the order.
21004	CancelReason	Y	Reason code for order cancellation. This is a custom FIX tag.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
1	Account	Y	Account associated to the order.
67	ListSeqNo	Y	As stated in the cancel request
21023	OrigListSeqNo	Y	Original list sequence number
	Standard Trailer	Y	

9.2.5 OrderCancelReject

The order cancel reject message is used by the exchange to indicate a cancel or a cancel/replace request cannot be honored.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 9
37	OrderID	Y	Unique identifier for an Order as assigned by the exchange. If the ClOrdID on the CancelRequest is unknown this field will contain "NONE".
11	ClOrdID	Y	Unique identifier of the cancel or cancel/replace request being rejected. This field is restricted to the use of uppercase alphanumerics.
54	Side	Y	As stated in the cancel request.
41	OrigClOrdID	Y	As specified on cancel/replace request.
39	OrdStatus	Y	The status of the order after this reject was applied.
434	CxlRejResponseTo	Y	Identifies the type of request that a Cancel Reject is in response to.
102	CxlRejReason	Y	Code to identify reason for cancel rejection.
	Standard Trailer	Y	

9.2.6 Restatement

The Restatement execution report is an unsolicited event to notify the client that an open order has been updated by the Trading System. This message will be sent for the order quantities canceled because of Match Trade Prevention (MTP).

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = 8
37	OrderID	Y	Unique identifier for an Order assigned by the exchange.
11	ClOrdID	Y	The ClOrdID associated with the order.
17	ExecID	Y	Unique identifier of execution message as assigned by the exchange. Note: the ExecID can be correlated with the OrderID fields in the Memoir feed messages.
150	ExecType	Y	D - Restated
39	OrdStatus	Y	0 - New 1 - Partially filled 4 - Canceled
21035	OptionsSecurityID	Y	The MEMX OptionsSecurityID that identifies the options security.
55	Symbol	Y	The OSI Root for the options security.
541	MaturityDate	Y	Date of maturity in YYYYMMDD format.
201	PutOrCall	Y	Indicates whether the option security is a put or a call.
202	StrikePrice	Y	The Strike Price for the options security.
54	Side	Y	As stated in the order.
151	LeavesQty	Y	Quantity open for execution.
14	CumQty	Y	Total quantity filled on the order.
44	Price	N	As stated in the order.
38	OrderQty	Y	The quantity after the restatement.
31	LastPx	Y	ExecRestatementType(3):The updated booked price. This is the price that the order can match

Tag	Field Name	Req'd	Meaning/Values
			at. It may be different than the displayed price. ExecRestatementType(5): The match price if this order had not been prevented from executing due to MTP restrictions.
32	LastQty	N	For ExecRestatementReason (5): The quantity that would have matched if this order had not been prevented from executing due to MTP restrictions.
378	ExecRestatementReason	Y	See ExecRestatementReasonType
21022	ExtendedRestatementReason	N	This field provides more RestatementReason info when applicable/available.
1	Account	Y	Account associated to the order.
60	TransactTime	Y	UTC timestamp since epoch with millisecond precision.
	Standard Trailer	Y	

9.3 AllocationInstructionAlert

The AllocationInstructionAlert message provides an unsolicited notification of allocations applied to a previously reported Trade or Allocation. This message is sent to a customer when a new Allocation Instruction has been applied to a Trade or Allocation.

Tag	Field Name	Req'd	Meaning/Value
	Standard Header	Y	With tag 35 (MsgType) = BM
70	AllocID	Y	Unique identifier for this allocation instruction ack message - generated by MEMX
626	AllocType	Y	As stated on request
71	AllocTransType	Y	Identifies allocation transaction type being requested (New, Replace, or Cancel)
72	RefAllocID	C	Required when AllocTransType = Replace, contains the IndividualAllocID of the allocation unit being changed.
796	AllocCancReplaceReason	C	Required when AllocTransType = Cancel
54	Side	Y	Side of the Execution
21035	OptionsSecurityID	Y	The OptionsSecurityID used to identify the options product for the order.
55	Symbol	Y	The OSI Root used to identify the options product for the order.
541	MaturityDate	Y	The maturity date used to identify the options product for the order.
201	PutOrCall	Y	The put or call indicator used to identify the options product for the order.
202	StrikePrice	Y	The strike price used to identify the options product for the order.
75	TradeDate	Y	YYYYMMDD
124	NoExecs	Y	Number of Trades being allocated (always 1)
→ 1003	Tradeld	Y	Specifies the TrdMatchId of the original trade being allocated
→ 32	LastQty	Y	Specifies the LastQty of the original trade being allocated
→ 31	LastPx	Y	Specifies the LastPx of the original trade being allocated

Tag	Field Name	Req'd	Meaning/Value
78	NoAllocs	Y	The dimensions of the Allocations Repeating Group.
→ 80	AllocQty	Y	The quantity of the individual allocation being made
→ 1047	AllocPositionEffect	N	The position effect of the individual allocation being made
→ 467	IndividualAllocID	Y	The unique identifier for this specific allocation unit, used to identify this allocation in future updates
→ 539	NoNestedPartyIDs	Y	The number of nested parties in the following NestedParties group
→→524	NestedPartyID	Y	
→→525	NestedPartyIDSource	Y	
→→538	NestedPartyRole	Y	
	Standard Trailer	Y	

9.4 UserNotification

The UserNotification message is used to notify the user of an event or information about the exchange. This message will be used to provide a logical session shutdown warning due to an up and coming maintenance window.

Tag	Field Name	Req'd	Meaning/Values
	Standard Header	Y	With tag 35 (MsgType) = CB
926	UserStatus	Y	Reason for notification.
	Standard Trailer	Y	